# OfficeLink

Complete application-building prompt

Act as a senior Python desktop developer and software architect. Build a complete, working Windows application called **OfficeLink** for **R & A Group**, an office with approximately 20 employees.

The application must provide internal employee messaging and document sharing through the office network. Deliver the actual source code, installation files, documentation and tests - not only a proposal or interface demonstration.

## 1. Business requirements

- Employees must communicate directly with each other, including short messages such as "Please come to my office."
- Employees must be able to send and receive documents.
- Each employee must have an individual username and password.
- The application must operate independently of Zoho Mail, Microsoft Teams, email accounts and telephone numbers.
- Organise communication around employees, rather than clients, audit assignments or projects.
- Access is required only within the office LAN or office Wi-Fi. Remote access from outside the office is outside the scope.
- Normal operation must not require internet access after installation.

## 2. Architecture and technology

Use a central host-and-client architecture:

- One Windows computer runs the central server.
- Employee computers run the desktop client.
- Store accounts, conversations, messages and attachment information centrally.
- Use Python with a professional PyQt6 or PySide6 interface.
- Provide an entry point named `OfficeLink.py` that can run from Windows IDLE by pressing F5.
- Keep the implementation modular and maintainable. Explain that the complete application folder must be retained.
- Use SQLite initially, with safe transactions, concurrent-access handling and consistent backups.

The initial host address is **192.168.70.122**, using **HTTPS on port 8443**. Make both configurable; do not hard-code this address throughout the application.

## 3. Employee application

Provide:

- Individual sign-in and logout.
- An employee directory showing display names and online/offline status.
- Direct one-to-one conversations.
- Persistent chat history with timestamps.
- Unread-message indicators.
- Desktop notifications and optional notification sounds.
- A clear distinction between sent messages and failed/pending messages.
- Document attachment, download and save functions.
- Clear connection status and useful error messages.
- Safe recovery after a temporary network interruption, without duplicating messages.
- Password changes.
- Minimise-to-tray support, with an explicit Quit option.

The interface should use a clean, professional light theme with navy and teal accents, readable fonts, an employee list on the left and the selected conversation on the right.

Voice calls, video calls and client/project management are not required.

## 4. Document sharing and storage

- Set the shared attachment capacity to **5 GiB for the whole office**, clearly labelled in the interface.
- Set the default maximum individual attachment size to **50 MiB**.
- Make these limits centrally configurable.
- Display used and available attachment capacity.
- Support PDF, DOCX, XLSX, PPTX, CSV, TXT, PNG, JPG, JPEG and ZIP files.
- Enforce file-size, storage and permission checks on the server.
- Use safe stored filenames and prevent path traversal.
- Ensure failed or interrupted transfers do not create corrupted documents or incorrect storage totals.
- Avoid excessive memory consumption during transfers.
- Allow only authorised conversation participants to download its attachments.
- Do not automatically delete messages or documents by default. Any retention policy must require an explicit administrator setting and clearly explain its effect.

## 5. Administrator console

Provide a separate host-management interface with:

- **Set up a new office**.
- **Use an existing host setup**.
- **Move office from another computer**.
- Start and stop host.
- Add employee.
- Reset password.
- Disable employee account.
- View employee accounts and their status.
- Create a staff installation ZIP.
- Back up the database.
- Prepare automatic Windows startup.
- Show server address, data location, storage usage and operating status.

Account creation should use temporary passwords that employees must change at first sign-in. Prevent accidental disabling of the last administrator.

Protect administrative access and document the role of Windows account and folder permissions.

## 6. Security and data protection

Implement:

- Password hashing using an established password-hashing method, with unique salts.
- No plaintext password storage or password logging.
- Login attempt throttling and session expiry.
- Server-side authorisation for every conversation and document request.
- HTTPS with certificate and hostname verification enabled.
- A documented method for generating, distributing and renewing the required certificates.
- Private office-network binding and a narrowly scoped Windows Firewall rule.
- Protected storage for the database, private certificate keys and backups.
- Logs suitable for troubleshooting without recording passwords, tokens or document contents.

Do not disable certificate checks or Windows Firewall to resolve connection problems. Do not claim that TLS provides end-to-end encryption.

## 7. Installation and desktop shortcuts

Make installation straightforward for a nontechnical administrator:

- Provide `Start_OfficeLink.cmd`.
- Check dependencies before importing external libraries.
- Install missing dependencies into an isolated environment, with visible progress and understandable failure messages.
- Do not silently upgrade a working environment.
- Provide one fallback `python -m pip install ...` command covering all external dependencies.
- Clearly distinguish installing Python from installing Python libraries.
- Automatically create a desktop shortcut after successful setup.
- Provide a separate shortcut-repair command.
- Handle paths containing spaces and redirected desktop folders.
- Generate a staff ZIP containing only the client application and public connection settings.

Never include the live database, private certificate keys, administrator tools or passwords in a staff ZIP.

If you provide an executable installer, identify what was actually built and tested. Do not label Python source files as a compiled installer.

## 8. Automatic startup and reliability

Provide a supported Windows background-hosting method that:

- Starts after a reboot without requiring an interactive Windows sign-in.
- Does not require the administrator console to remain open.
- Uses appropriate account permissions.
- Supports controlled restart after failure.
- Prevents duplicate server processes from using the same database.
- Supports graceful shutdown and useful operational logs.

Document the local installation steps and a reboot test from an employee computer. Verify operating-system and runtime compatibility before recommending deployment.

## 9. Backup, recovery and updates

Provide:

- Consistent database backups containing users, chats and stored attachments.
- Instructions for separately backing up configuration, certificates and recovery information.
- A restore procedure tested on synthetic data.
- A migration procedure that preserves existing accounts and history.
- Source-code updates that do not overwrite live data or configuration.
- A rollback procedure.
- A clear explanation that a source-code ZIP alone is not a backup of live office data.

Never initialise a new office automatically when an existing database cannot be opened.

## 10. Testing and deliverables

Test meaningful scenarios, including:

- Multiple simultaneous employee connections.
- Direct messaging and persistence after restarting.
- Document upload/download and unauthorised-access rejection.
- File-size and shared-storage limits.
- Account disabling and password resets.
- Network interruptions.
- Backup and restore.
- Duplicate-host prevention.
- Desktop shortcuts and automatic startup, where a Windows environment is available.

Clearly separate automated tests, simulated checks and tests performed on actual Windows computers. Do not claim checks that were not executed.

Deliver:

1. Complete editable source code in a downloadable ZIP.
2. Host and staff setup instructions.
3. Dependency requirements.
4. Desktop-shortcut support.
5. Automatic-startup instructions.
6. Backup, restore and migration instructions.
7. Troubleshooting guidance for common errors.
8. A source-file map explaining each module.
9. Test results and remaining deployment checks.

Use reasonable defaults and proceed with implementation. Ask questions only where missing information materially affects the design. Prioritise a reliable, understandable application that office staff can install and use.